

MATH 4790: Quiz 4

March 30, 2026

Name: _____

Score: _____/15

Instructions: Show all work and justify your answers. Only calculators approved for Exam FM are permitted. Credit will not be given to just guessing the correct answer.

Question 1.

A loan is repaid with 18 annual level payments made at the end of each year. The annual effective interest rate is 5.4%.

The interest portion of the second payment is equal to the principal portion of the t th payment.

Find t .

- (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

Question 2.

An office takes out a \$1,250 equipment loan at time $t = 0$, where time is measured in years. The annual effective interest rate is 20% for the first two years, and 25% for all years thereafter.

The loan is scheduled to be repaid using the following payments:

- a payment of X at time $t = 2$,
- a payment of Y at time $t = 4$, and
- a final payment of 625 at time $t = 5$.

No other payments are made.

At time $t = 3$, the outstanding balance on the loan is 1,000.

Find $\frac{X}{Y}$.

- (A) $\frac{5}{4}$ (B) $\frac{4}{3}$ (C) $\frac{3}{2}$ (D) 2 (E) $\frac{9}{4}$

Question 4.

A 20-year loan is repaid with end-of-month payments. Let j denote the nominal annual interest rate compounded monthly.

For the first year, payments are \$300 each. The remaining payments are \$500 each.

Right before the 12th payment, the outstanding loan balance is \$28,000.

Find j .

- (A) 19.5% (B) 20.1% (C) 20.7% (D) 21.3% (E) 21.9%